

AXI4-Lite Firewall IP Core

Block Diagrams and Descriptions

IP core	<code>altera_axi4_lite_firewall</code>
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Target	Intel FPGA / Quartus Prime, Platform Designer

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1. System context

Where the core sits, and what must be connected to it.

The core is a bump-in-the-wire between one master and one protected peripheral. It enforces an allow-list on every transaction and guarantees the master always gets a response, even when the peripheral stops answering.

Each transaction's address is compared against a table of ranges, each with independent read and write permission. The lowest-index valid rule containing the address wins; ranges need not be disjoint, so put more specific rules at lower indices. Default-deny applies:

Condition	Response
No rule matches	DECERR
Rule matches, wrong direction	SLVERR
Blocked while isolated	SLVERR

A denied read returns zeroed `RDATA` rather than stale bus data, so a rejected read cannot leak the result of an earlier permitted one.

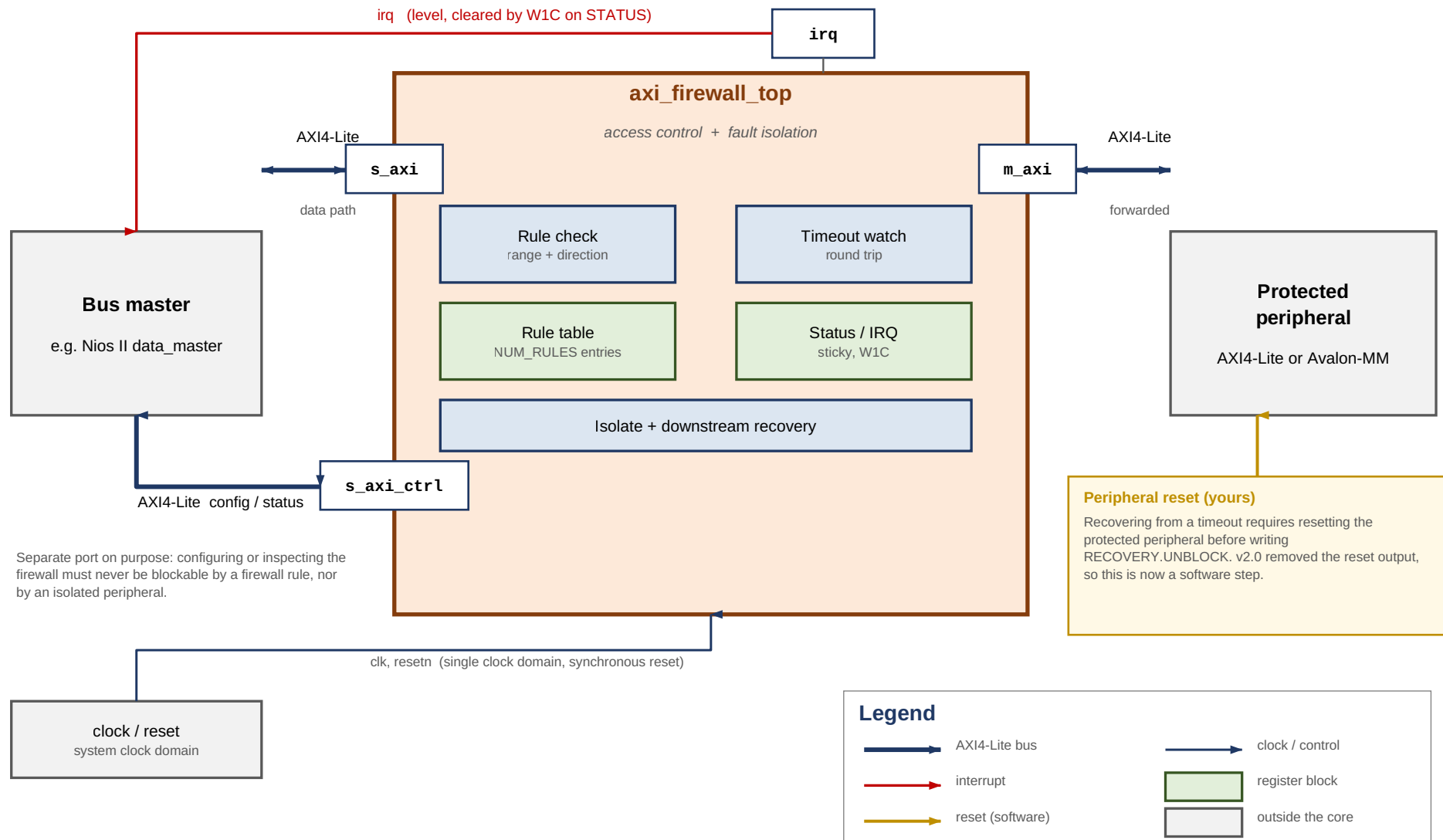


Figure 1. System Context

1.1 Fault isolation

Every forwarded transaction is watched by a timeout covering the whole round trip, address issue to response. That catches a peripheral that never raises `AWREADY` / `ARREADY` as well as one that accepts and then goes quiet. On expiry the core answers `SLVERR` upstream immediately, so the master never hangs, and latches an internal broken state that blocks all further forwarding until software acknowledges the fault.

1.2 Why the control port is separate

`s_axi_ctrl` is a physically distinct AXI4-Lite slave. Recovery requires writing `STATUS`, and if that write had to pass through the firewall's own rule check, or through an isolated peripheral, a fault would be unrecoverable. Both ports may be driven by the same master.

1.3 Recovery is a software sequence

```
ack the fault → poll the busy bits (bounded) → RESET THE PERIPHERAL
              → write RECOVERY.UNBLOCK
```

Caution

`UNBLOCK` is what withdraws a stuck `VALID`. Skipping the reset makes that a protocol violation on a live bus: measured, 1 of 25 timing offsets then mis-attributes a stale response, against 0 of 25 when the sequence is followed.

Version 2.0 removed the `m_axi_resetsn` output the core used to drive, so resetting the protected peripheral is now the integrator's job. It must be reachable from software — a Platform Designer reset bridge under software control, or a PIO output, both work.

1.4 Interface summary

Interface	Connects to
<code>s_axi</code>	The master being policed
<code>m_axi</code>	The protected peripheral
<code>s_axi_ctrl</code>	Rule table, status, IRQ enable
<code>irq</code>	A CPU interrupt input
<code>clk</code> , <code>resetsn</code>	Single clock domain, synchronous reset

Note

There is no per-master filtering. AXI4-Lite carries no ID field, so the core cannot tell one master from another. It polices *what* is accessed, not *who* is accessing it.

2. Internal architecture

Inside `axi_firewall_top`: two independent datapaths, one register block, one recovery controller.

The datapaths are fully independent — separate FSMs, capture registers, lookup ports and fault pulses — so a read and a write can be in flight at once. Only `FAULT_ADDR` and `FAULT_INFO` are shared; if both fault in the same cycle the write side wins, a deterministic tie-break.

`chk_*_` is the rule-lookup bundle: `chk_*_addr` out to the register block, `chk_*_match` and `chk_*_allow` back, all combinational within one cycle. Each datapath has its own port, so neither ever waits on the other for a verdict.

Both datapaths follow the same four-state shape. IDLE waits for a request and captures it; EVAL applies the verdict in a single cycle; FWD drives the master side and runs the timeout counter; RESP holds the response until the master accepts it.

EVAL is where policy is decided, in strict priority order:

1. Forwarding blocked → `SLVERR`
2. Global bypass (`CTRL.GLOBAL_ENABLE` clear) → forward unconditionally
3. No rule match → `DECERR`
4. Rule matched but direction denied → `SLVERR`
5. Otherwise → forward

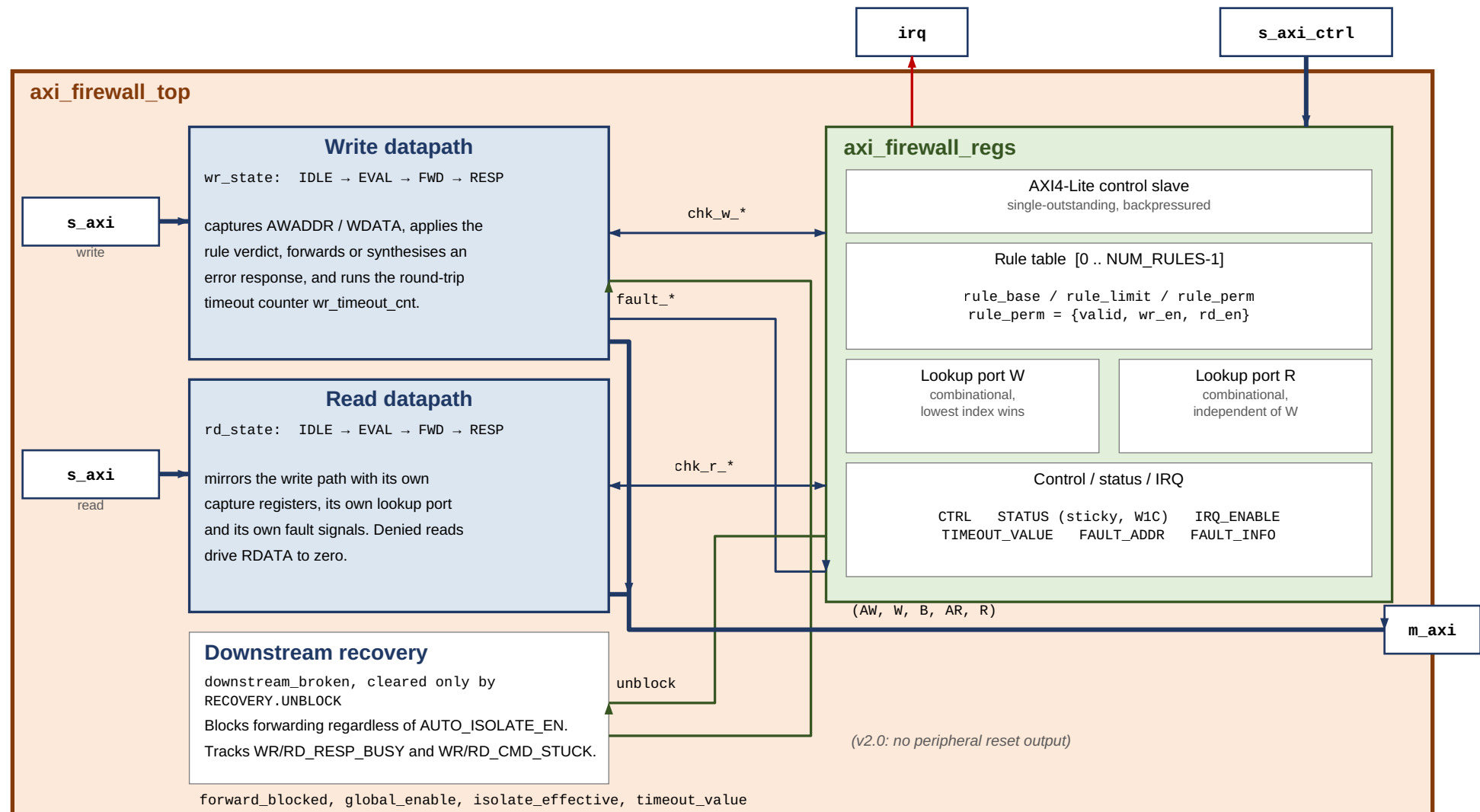


Figure 2. Internal Architecture

2.1 The register block

`axi_firewall_regs` owns the rule table and all software-visible state, and exposes two independent purely combinational lookup ports so both datapaths get an answer in the same cycle without contending.

The lookup is a priority chain over `NUM_RULES` entries. It is the likeliest critical path, and the standard fix if it limits f_{MAX} is to register it with an extra pipeline stage.

2.2 Timeout and recovery

On expiry the core reports `SLVERR` upstream, latches `downstream_broken`, and deliberately leaves any asserted `m_axi_*VALID` asserted.

Caution

AXI requires `VALID` to hold until `READY`; withdrawing it can wedge the interconnect between the core and the peripheral, not just the peripheral.

The stuck `VALID` is dropped only by `RECOVERY.UNBLOCK`, which means software has reset the peripheral and its AXI state is gone. `STATUS` exposes `RESP_BUSY` (the peripheral owes a response) and `CMD_STUCK` (it never accepted the command) so a driver can sequence this.

Bound the busy poll. A peripheral that accepted a command and then died owes a response forever, so an unbounded poll hangs precisely when recovery matters most. A transaction arriving while blocked is rejected, not stalled, so drivers need a retry path.

`downstream_broken` is independent of `CTRL.AUTO_ISOLATE_EN`. That bit governs only the visible `ISOLATED` status; blocking after a timeout is required for protocol safety and happens either way.

2.3 Key internal signals

Signal	Meaning
<code>chk_w_addr / chk_r_addr</code>	Address presented to the rule lookup, one port per datapath
<code>chk_*_match</code>	Some valid rule contains this address; low means <code>DECERR</code>
<code>chk_*_allow</code>	That rule permits this direction; low means <code>SLVERR</code>
<code>global_enable</code>	<code>CTRL</code> bit 0. Low is bypass: forward everything unchecked
<code>isolate_effective</code>	<code>MANUAL_ISOLATE</code> or the auto-isolate latch
<code>downstream_broken</code>	Latched by a timeout, cleared only by <code>RECOVERY.UNBLOCK</code>
<code>forward_blocked</code>	<code>isolate_effective</code> $\vee$ <code>downstream_broken</code> — the EVAL gate
<code>unblock</code>	Single-cycle pulse from a write to <code>RECOVERY</code>

3. Datapath state machines

Both FSMs are enum-typed, so Questa names the states in its coverage report. All 14 transitions are covered by the test suite.

IDLE waits for a request. The write path needs both `AWVALID` and `WVALID` before asserting either `READY` — a slave may always add wait states, so this is compliant and keeps one simple pattern. Address, protection bits, data and strobes are captured on entry to `EVAL`.

EVAL is always one cycle. The rule lookup result for the captured address is already available combinationally, so the verdict costs no extra cycle. If the downstream is blocked the answer is `SLVERR` — v2.0 removed the reset pulse and with it the window in which arrivals were stalled instead.

FWD drives the master side and runs the timeout counter. The counter starts when forwarding starts and covers the whole round trip, so a peripheral that never raises `AWREADY` is caught by the same mechanism as one that accepts and then goes silent.

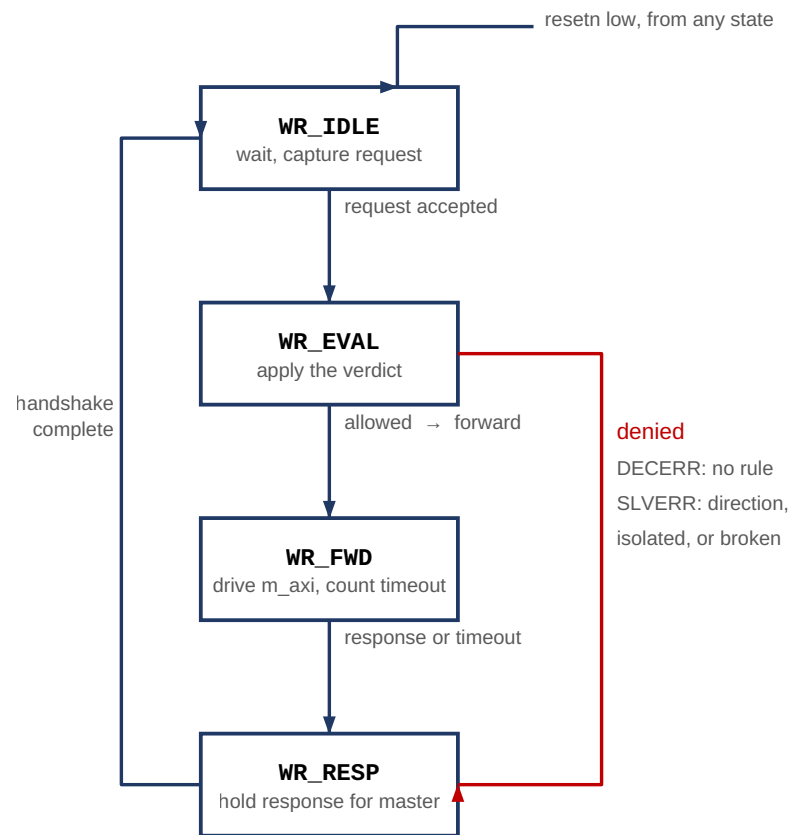
RESP asserts the response and holds it, with the payload stable, until the master's `READY` arrives.

The write and read machines are structurally identical but fully separate, each with its own capture registers and lookup port. A denied read drives `RDATA` to zero before responding, so it cannot return stale data from an earlier permitted read.

Caution

On a timeout, `m_axi_*VALID` is deliberately **not** withdrawn. The FSM leaves `FWD` and answers the master, but the master-side `VALID` stays asserted until software writes `RECOVERY.UNBLOCK` — the one point where dropping it is legitimate, because software has just reset the peripheral and its protocol state no longer means anything.

Write datapath (wr_state)



Read datapath (rd_state)

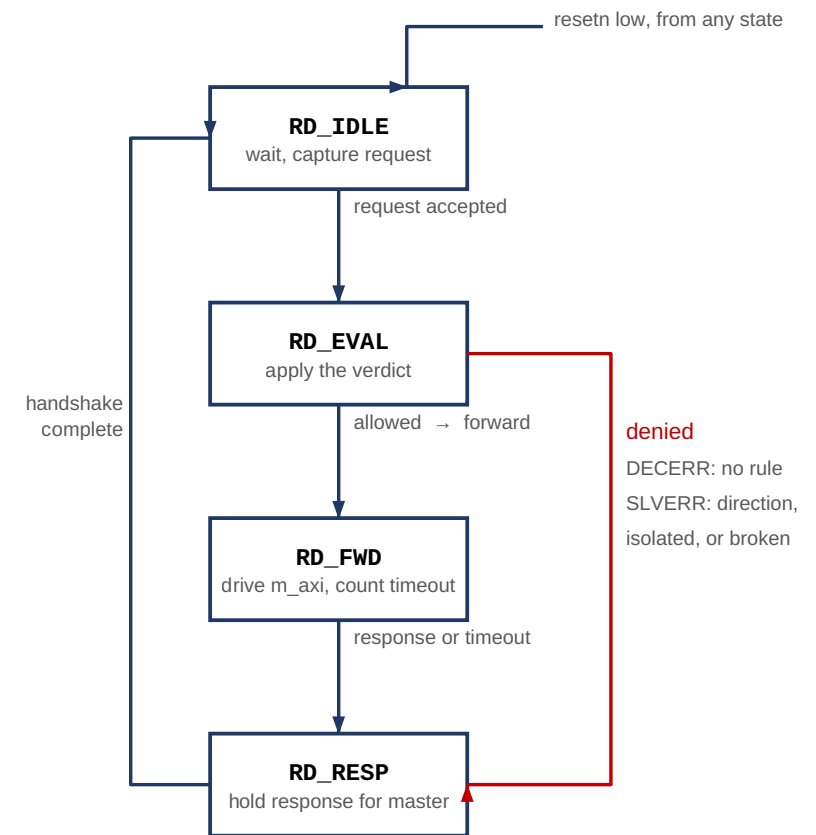


Figure 3. Datapath State Machines

3.1 Response encoding

Condition	Response
Allowed, peripheral answers	As returned by the peripheral
No valid rule matches	DECERR
Matched, direction denied	SLVERR
Blocked while isolated	SLVERR
Peripheral timed out	SLVERR

A transaction blocked because the core is isolated returns **SLVERR** but sets no status bit and raises no interrupt. Only genuine violations and timeouts are logged, so a burst of rejected traffic during isolation cannot bury the fault that caused it.

3.2 Latency

Measured against a zero-wait-state peripheral, counting clock edges from request assertion to response valid:

Operation	Cycles
Single write	6
Single read	6

The suite fails if either exceeds 8, so the figure cannot silently rot. Cost is per transaction and does not amortise — the core is single-outstanding and non-pipelined.

3.3 FSM transition coverage

Questa 2024.1, from the version 2.0 regression.

Transition	Write FSM	Read FSM	Exercised by
IDLE → EVAL	23	23	Every transaction reaching the core
EVAL → FWD	17	16	Every permitted transaction
EVAL → RESP	5	6	Denied, out-of-range and blocked transactions
EVAL → IDLE	1	1	Reset asserted during EVAL
FWD → RESP	14	13	Normal completion and timeout
FWD → IDLE	3	3	Reset asserted during FWD
RESP → IDLE	19	19	Every completed response

4. Register map and rule table

All registers are 32 bit and word aligned on `s_axi_ctrl`. Reset values are shown; the core comes up secure by default.

4.1 Fixed registers

Offset	Name	Access	Reset
0x00	CTRL	R/W	0x3 (enabled, auto-isolate on)
0x04	STATUS	R / W1C	0x0
0x08	IRQ_ENABLE	R/W	0x7 (all enabled)
0x0C	TIMEOUT_VALUE	R/W	all ones (effectively disabled)
0x10	FAULT_ADDR	R	0x0
0x14	FAULT_INFO	R	0x0
0x18	CORE_INFO	R	version (0x0200) and NUM_RULES
0x1C	RECOVERY	W	bit 0 UNBLOCK , self-clearing

4.2 Rule table

Offset	Name	Description
0x40 + i·0x10	RULE_BASE[i]	Inclusive base address
0x44 + i·0x10	RULE_LIMIT[i]	Inclusive top address
0x48 + i·0x10	RULE_PERM[i]	Permissions and valid bit

`i` runs 0 to `NUM_RULES - 1`. With the default 8 rules the table spans 0x40–0xBF. `CTRL_ADDR_WIDTH` must be wide enough to reach the whole table; a validation callback in `axi_firewall_hw.tcl` enforces this, because an undersized control port silently makes high-index rules unreachable.

4.3 Bit fields

CTRL (0x00)

31:3 reserved	2 MANUAL_ISOLATE	1 AUTO_ISOLATE_EN	0 GLOBAL_ENABLE
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STATUS (0x04) - bits 2:0 sticky, write 1 to clear

8:7 RDWR_CMD_STUCK	6:5 RDWR_RESP_BUSY	4 BLOCKED	3 ISOLATED	2:0 the three sticky bits
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FAULT_INFO (0x14)

31:4 reserved	3:1 type: 1=ADDR 2=PERM 3=TIMEOUT	0 WAS_WRITE
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RULE_PERM[i] (0x48 + i·0x10)

31:3 reserved	2 VALID	1 WRITE_ALLOW	0 READ_ALLOW
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CORE_INFO (0x18)

31:16 version 0x0200 = v2.0	15:8 reserved	7:0 NUM_RULES
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Figure 4. Register Bit Fields

4.4 Programming model

Bring-up is: program the rules, set a timeout, enable interrupts. The core is secure by default — `GLOBAL_ENABLE` resets to 1 and the rule table resets empty, so with no configuration at all every transaction is denied rather than passed.

Note

`TIMEOUT_VALUE` resets to all ones, which is effectively no timeout. Set it to a real round-trip budget for your clock and peripheral, or fault isolation will not trigger in any useful time.

Reconfiguring a live rule

A rule is three registers, so updating one that is currently `VALID` leaves a window where `BASE` is new but `LIMIT` is still old. Clear `VALID` first — unnecessary at init, since `VALID` is 0 out of reset:

```
write RULE_PERM[i] = 0
write RULE_BASE[i], RULE_LIMIT[i]
write RULE_PERM[i] = perms | VALID
```

4.5 Interrupt handling

`irq` is a level interrupt, asserted while any enabled sticky `STATUS` bit is set. Clear at the source by writing 1 to the relevant bit; there is no separate acknowledge register. This is the standard memory-mapped-peripheral idiom and works directly with the Nios II HAL ISR pattern.

`FAULT_ADDR` and `FAULT_INFO` capture the most recent fault of any type. If a read and a write fault in the same cycle, both sticky bits set correctly but the capture registers take the write side.

Caution

Clearing `TIMEOUT_ERROR` is not enough on its own. It also releases the auto-isolate latch. What it does **not** do, as of v2.0, is reopen the downstream — that takes an explicit `RECOVERY.UNBLOCK` after software has reset the peripheral. A v1.x driver stops here and then sees every access return `SLVERR`.

4.6 Control port behaviour

`s_axi_ctrl` is single-outstanding and backpressures correctly: `AWREADY` and `WREADY` are withheld while a `BVALID` is unacknowledged, and `ARREADY` while an `RVALID` is. A driver that issues one access at a time sees no difference; one that pipelines simply waits.

Before v1.2 the port asserted `READY` on `VALID` arriving without checking whether the previous response had been accepted. A pipelined second access got its handshake taken, was silently dropped, and never produced a response. Measured before the fix: two writes produced one response. After: two for two.

5. Verification status

Questa 2024.1, Verilator 5.48 and slang 11, against the version 2.0 sources.

Metric	Result
Self-checking tests	103 / 103
Assertions	14 / 14
Cover directives	6 / 6
FSM states	8 / 8
FSM transitions	14 / 14
m_axi protocol violations	0

Every assertion has a non-zero non-vacuous pass count.

5.1 Not yet verified

Quartus analysis of the `_hw.tcl` component, synthesis results (logic element count, f_{MAX}), and behaviour inside a generated Platform Designer interconnect. See section 8.4 of the [user guide](#) for the full list.

The rule-table decode is checked by lint across nine parameter configurations, which proves it elaborates — not that it works. That is exactly where an off-by-one would hide, which is why the functional suite programs and reads back every rule slot.